Classification scheme

Data extraction form for the included studies

Telmo Miguel-Medina

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One row per included study, filled in from the full text.

Codes: y yes, p partial, n no, ns not stated. Lists are separated with semicolons. Every y or p in block K is backed by the literal sentence and the page where it appears.

1. Extraction form

Block	Field	Values
A. Context	Reference, DOI, research group	text
	Level	integrated circuit, board, both, ns
	Stage	production, maintenance, in-field, design, ns
	Validation	simulation, hardware, both
	Circuits; size of the largest	list; number of components
B. Selection	Selects nodes / stimuli / measurements	y, n (one field each)
	Type of stimulus	DC, AC, transient, other
	Coupling of the decisions	joint, sequential, single
	Number of candidate tests	integer or ns
C. Faults	Fault type	catastrophic, parametric, both
	Multiple faults; number of faults	y, n; integer or ns
D. Objective	Objective	detection, isolation, identification, specification compliance
	Multi-objective	y, n
	Weighting by fault probability	y, p, n
	Weighting by fault severity	y, p, n
	False alarms controlled	y, p, n
E. Cost	Cost model	none, count of tests, abstract, time, hardware
	Model of the instrument (noise, resolution, range)	y, p, n
	Node accessibility considered	y, n
F. Tolerances	Treatment of tolerances	ignored, threshold, parametric, Monte Carlo
	Monte Carlo runs per fault; distribution	integer or ns; normal, uniform, other, ns
	Confidence intervals reported	y, n
G. Algorithm	Family	exhaustive, greedy, graph search, metaheuristic, exact, machine learning, reinforcement learning, symbolic, other
	Optimality or approximation guarantee	y, n
H. Comparison	Compared with an exact optimum	y, n
	Compared with the ambiguity-group limit	y, n

Block	Field	Values
	Baseline methods	list
J. Reproducibility	Netlist given; fault list given; parameters given	y, p, n (one field each)
	Code; data	URL or n
K. Cross-cutting criteria	C1, C2, C3, C4, C5a, C5b	y, p, n (defined below)
	Notes	text

2. Cross-cutting criteria

They summarise blocks B to J into six properties, so that the studies can be compared in a single matrix.

Id	Criterion	“Yes” means
C1	Joint selection	At least two of {node, stimulus, measurement} are decided in the same optimisation
C2	Hardware cost	The cost distinguishes physical resources, or the model includes the limits of the instrument
C3	Risk	Faults are weighted by probability or severity, or false alarms or test escapes are constrained
C4	Statistical certification	Confidence intervals are given, or the sample size is justified
C5a	Reference	The result is compared with an exact optimum or with the limit set by the ambiguity groups
C5b	Openness	Code or data sufficient to repeat the experiment are published

“Partial” is used when the property is present but incomplete; for example, C2 when the cost is the test time on a production tester, or C4 when the number of Monte Carlo runs is discussed but no interval is given.

3. Published figures

A separate table, one line per study and circuit, records the results reported for the most common benchmark circuits, so that they can be compared across studies.

Fields: study, circuit, tests selected, number of tests, metric, value, table or figure of origin.